



Last Mile Delivery in Hammersmith and Fulham: An Alternative Approach



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Introduction

Being one of the first London boroughs to declare a climate emergency in 2019, Hammersmith and Fulham (H&F henceforth) council have identified the opportunity to further their commitment to the climate and the wider H&F community via assessing the issue of fossil-powered vehicles for last mile delivery. Drawing upon existing examples through a case study approach, this report aims to present an alternative approach to last-mile delivery, focusing on the provision of low-carbon last-mile delivery for all businesses, households, and individuals throughout the borough. We aim to present a vision for last mile delivery that goes beyond sustainable and effective parcel delivery. Our vision of Parcels, People, and Planet, aims to directly benefit all those involved with the last-mile delivery process, and promotes a gradual decoupling from the wider global supply chain, through prioritising shopping local and encouraging a wider-sense of community. Based upon global examples and building upon research carried out within the borough itself, this report will present a last-mile delivery system based around three types of urban logistics hubs, medium-sized hubs, small-sized hubs, and community-facing hubs. In considering the real-world implementation of this system for H&F council, accompanying policy proposals and strategic recommendations will also be presented.

1. Last Mile Delivery in H&F

1.1. Understanding Last Mile Delivery

In the wake of the Covid-19 pandemic, the existing shift in consumer behaviour from shopping locally to shopping online has been accelerating (ONS, 2020). This trend, and the accompanying increase in parcels being shipped worldwide has further highlighted the last step of the delivery process, i.e. that of last mile delivery.

In this report, last mile delivery will refer to all logistical activities related to the last stage of parcel transportation, i.e. from when a parcel is processed at its final depot and ending with the parcel being successfully received by the recipient (Boysen et al, 2021). As is the case with much of the existing global logistics system, the last mile of parcel delivery is carbon intensive, predominantly relying on the use of vans and cars, the vast majority of which are powered by fossil fuels, negatively contributing to climate change (United Nations, n.d.).

1.2. Overview

Within dense urban areas, such as H&F, the existing fossil-fuel powered approach to last-mile delivery has a significant impact on the day-to-day lives of residents and on the planet. Addressing the use of fossil-fuel powered last mile delivery is crucial in achieving H&F councils' vision for net zero carbon emissions by 2030, as set out in H&F 2030: Climate and Ecology Strategy (2021). Additionally, the number of parcels being delivered within London is set to double by 2030, further indicating a need for change (Centre for London, 2021). Within H&F three main issues with the existing fossil-fuel oriented approach to last-mile delivery have been identified: increased air pollution, increased congestion, and ineffective parcel delivery.

1.3. Problems with Last Mile Delivery in H&F

Increased Air Pollution:

Within H&F, a commission carried out by the council found air pollution to be one of the main concerns for residents (H&F, 2021). Throughout London, freight and deliveries account for a quarter of London's total carbon emissions from transport (Centre for London, 2021). Additionally, fossil-fuel powered Light-goods vehicles (LGV's), the primary mode of transport for last-mile delivery, contribute around 13% of the transport sector's nitrogen oxide levels and 15% of the sector's particulate matter released within the city (ibid). Higher levels of nitrogen dioxide and particulate matter have been linked to a myriad of adverse health effects, particularly in vulnerable groups such as the elderly, the young, and those with pre-existing respiratory problems (UK Government, 2023).

Increased Congestion

Within inner boroughs like H&F, congestion presents a significant problem, with the borough experiencing the highest level of congestion within West London (H&F Council, 2011). Due to the highly competitive nature of last-mile delivery, delivery vans compete for space on central and residential roads, further contributing to the existing level of congestion within the city (Centre for London, 2021). In 2021, vans accounted for 16% of motor traffic in London, and current trends show a steady increase in the miles travelled by LGV's on London roads (Allen, 2018a). Also, it is estimated that within London parcel vans spend around 4 hours parked each day, often taking up parking spaces and space on pavements (Allen et al., 2018b). Not only does this take space away from other road users such as cyclists and pedestrians, but these vans are often left on when parked, further contributing to air pollution (Chaney et al., 2017).

Ineffective Parcel Delivery

The ubiquitous nature of fossil-fuel delivery vans on London streets, in addition to the promises of next day delivery suggests the existing last-mile delivery system is not only convenient, but highly efficient for both customers and couriers. This is not necessarily the case. For example, despite fossil-fuel powered vans being the most common method of delivery, Allen et al (2018), found that walking accounts for 62% of parcel carriers' roles. Furthermore, research has shown that e-cargo bikes are able to carry out around 2.5 times more deliveries than vans within the same time frame (Reid, 2019). Additionally, increasingly strict delivery schedules from operators results in parcels being left outside, with a recent survey from Which, the consumer website, finding that more than a fifth of those surveyed experienced parcels being left, outside rather than handed directly to the resident (Sainato, 2021; Witherden, 2023).

1.4. Parcels not Pollution

Within H&F, positive action has already been taken to explore alternatives to fossil-fuel powered last-mile delivery. The Parcels not Pollution initiative was established from a partnership between Hammersmith Business Improvement District (BID), H&F council, and e-cargobikes.com, in addition to receiving support via funding from Transport for London. As indicated in the name, this initiative was launched to address most of the issues associated with air pollution and to improve air quality within Hammersmith town centre (Hammersmith BID, 2019). This initiative involves the use of the e-cagobike.com warehouse, which functions as a micro-logistics hub to which businesses could order parcels. These parcels are subsequently delivered to the businesses via e-cargo bike. Additionally, H&F council provides a free advice service for local businesses interested in becoming more sustainable, in addition to providing subsidies for local businesses to purchase their own cargo bikes. Originally beginning in 2019, the scheme has now been extended until April 2025.



Figure 1: Parcels Not Pollution cargo bike
Source: H&F Council (2024)

While the Parcels not Pollution initiative demonstrates the H&F council's commitment to creating a sustainable approach to last-mile delivery, numerous limitations have been identified.

Limitations:

- Delivery is currently limited to local businesses based within the town centre.
- While e-cargo bikes present a low-carbon alternative, highly variable weather conditions within the UK can limit their year-round effectiveness.
- Like many London boroughs, busy roads and limited cycle infrastructure can negatively impact the safety and efficacy of a cargo-bike only system.
- Despite aiding local business by providing a centralised alternative, this strategy does not actively work to incentivise local community engagement with said local business, or for local businesses to further collaborate with each other.

1.5. Our Vision and Goals

Considering the above challenges associated with Last-Mile Delivery within H&F, in addition to existing limitations associated with the Parcels not Pollution scheme, we have developed the following vision and goals for last-mile delivery within the borough. We have taken a three pronged approach, aiming to prioritise **Parcels, People, and Planet**.



Parcels

- Create a last-mile delivery system able to process parcels of various sizes, operate across a range of scales, and provide for all recipient types, e.g. businesses, households, and individuals.
- Focus on a system where parcels get to the recipient safely and effectively.



People

- Centre the needs and requirements of all people involved in the last-mile delivery system, including operators, workers, and users.
- Include the local community with H&F net zero 2030 vision by promoting active transport and shopping local



Planet

- Prioritise the use of sustainable, low-carbon modes of transportation to reduce air pollution
- Promote parcel delivery and collection methods via active transport through the integration of bike storage and repair facilities, EV charging stations, and inviting urban design.

1.6. Identifying Hub Requirements

As a result, our objective for this report is to conceptualise a last-mile delivery system based around urban logistics hubs across scales, increased utilisation of low-carbon transport such as elective LGVs, e-cargo bikes, cargo-bikes, and to facilitated increased usage of click and collect via personal bike or on foot.

Identifying User Requirements

In order for these hubs to be effective, the needs of all involved must be considered. For logistics hubs operating with H&F, the following requirements for urban delivery hubs have been identified:

Table 1: Summary of the different requirements of all those involved with the hubs.
Sources: Cross River Partnership (2020), Urberpeople.com (n.d.), Transport for London (2017; 2018)

Operator Requirements	Worker Requirements	User Requirements
Parking availability for both electric vehicles and cargo bikes	Toilets	Clearly marked and attractive hubs
Security Patrols	Hydration stations	Community Notice Board
Electric Vehicle charging station	Bike Repair	Street furniture
CCTV	First Aid	CCTV
Network Coverage 4G/5G Strong phone signal Wi-Fi	Network Coverage 4G/5G Strong phone signal Wi-Fi	Network Coverage 4G/5G Strong phone signal
Parcel Trolleys	Device charging facilities	Phone Charging facilities
24/7 Accessibility	Safe road signs and signals	Safe road signs and signals

Identifying Wider Needs

In addition to creating an effective and sustainable last-mile delivery system, we believe that the use of these hubs can also benefit the wider needs of the community and the goals of H&F Council. We aim for these hubs to have a social focus, promoting a sense of community and encouraging residents to meaningfully engage with each other and the local community. Additionally, building upon H&F Council's Transport Plan (2011), these logistics hubs will also serve to promote active transport, i.e. walking and cycling, with the council identifying a clear potential for the expansion of these transport modes, as demonstrated in the creation of the Hammersmith Cycle Hub.

Hammersmith Cycle Hub

Within the borough, the creation of the Hammersmith Cycle Hub provides an example of how effective use of space and provision of secure bike storage can work to promote the use of bikes for residents and commuters. This hub is centrally located and offers a high level of bike security. Using an app, residents can become a member of the hub and can use it free of charge after doing so. This hub also contains bike repair and maintenance facilities, which are also free to use. The success of the cycle hub can also provide insight into how to effectively implement infrastructure within the borough.



Figure 2: Hammersmith Cycle Hub
Source: H&F Council (2022)

Main Takeaways:

1. **Need for effective use of space:** Being an inner borough space is limited, the cycle hub is located under the flyover taking advantage of what would potential be an empty lot.
2. **Conveniently located for users:** Located near the major transport hubs of the bus and tube stations, as well as near the town centre.
3. **Clearly identifiable:** Clear branding and labelling, helps for new users to located and potential users to identify the name for later research.
4. **Multi-use:** Functions as both bike and as a bike repair facility.
5. **Effective use of Technology:** Corresponding roll out of an app to track usage and membership.

1.7. Case Study Approach

Building upon the outlined case studies from within H&F, the following section will explore and evaluate case studies from around the world, to further inform our vision and explore how an alternative approach to urban last-mile delivery can best to cater for the identified needs.

2. Review of Case Studies & Policies

2.1. Pimlico Logistics Hub in London

For last-mile delivery hubs, collaboration among stakeholders is crucial for providing top-notch services to customers and staying competitive in the market (Ko et al., 2020). A public-private partnership approach can effectively address the interests of all involved parties and ensure the project's continuity. The Pimlico Logistics Hub in London, led by the Cross River Partnership (CRP) in partnership with Q-Park, exemplifies a trial micro logistics hub. CRP, overseen by the Westminster City Council, operates as a public-private partnership, collaborating with entities such as the Westminster City Council, Victoria BID, and Transport for London within the Westminster Borough. This collaborative framework, engaging local authorities, private sector partners, and strategic agencies, expedites project implementation by incorporating diverse perspectives. Located at Q-Park UK Pimlico, the hub serves as a pioneering logistics solution and acts as a vital hub for the last-mile delivery operation. It functions as the primary point of package reception for residents of the Westminster area (Vollmer, 2023).

Upon consolidation, received parcels undergo further sorting and processing to ensure efficient dispatch. Subsequently, these packages are swiftly delivered to local residents via electric cargo bicycles operating in Westminster and surrounding areas, ensuring environmentally friendly and efficient distribution (CRP, 2023). Additionally, despite being situated within the residential parking area of Q-Park Pimlico, only six parking spaces have been leased for electric cargo car parking. The sorting of packages utilises other underutilised spaces within the parking facility to optimise space usage (Higgins, 2020). Furthermore, the loading and unloading of packages occur in other areas of level 0, away from the main entrance, to avoid disrupting other customers' regular use of the parking facility and to maintain noise levels at a minimum, showcasing a thoughtful approach to customer experience and facility management (CRP, 2023).



Figure 3: Pimlico Logistics Hub
Source: Cross River Partnership (2023)

This innovative logistics model not only provides residents with a more convenient and sustainable delivery service but also alleviates urban traffic congestion and reduces air pollution, reflecting the modern city's pursuit of green and intelligent logistics systems. According to the feedback report from the Micro Logistics Hub Trial in Pimlico, emissions have already been reduced by 184 kilograms of carbon dioxide, 384 grams of nitrogen oxides, and 17 grams of PM2.5 in the first month of hub operation (CRP, 2023).

Services in the hubs ensure safety and convenience for users. This includes CCTV actively recording and monitoring the area 24/7, as well as security patrols with personnel regularly inspecting and monitoring the premises (CRP, 2023). Additionally, help points equipped with emergency buttons allow individuals to request assistance or report emergencies in public areas. Basic facilities such as motorcycle parking, Tap & Go, free umbrella hire, toilets, and 4G Wi-Fi signal are provided for users' convenience (Q-park, no date). Furthermore, Automated External Defibrillators (AEDs) are available as life-saving devices to protect residents' health by delivering timely electric shocks to restore normal heart rhythm during sudden cardiac arrest (Q-Park, n.d.)

Main Takeaways:

1. **Collaborative Funding:** Partnerships between governmental bodies like WCC and CRP and private sector entities such as Infinium Logistics drive urban logistics innovation.
2. **Centralized Logistics Management:** Amenities like AED and CCTV surveillance streamline operations, making the hub a logistics coordination centre.
3. **Community Engagement:** Services like free umbrella hire and motorcycle parking foster goodwill and strengthen community ties.
4. **Emergency Preparedness:** Equipped with resources like jump leads, the hub ensures quick response in emergencies.
5. **Efficient Sorting and Dispatch:** The hub facilitates swift parcel sorting and processing for optimized urban logistics.
6. **Operational Independence:** Designed with a separate operational zone and storage company, the hub ensures smooth operations without disturbing nearby residents.

2.2. Finland's Parcel Locker Network

Background:

Parcel lockers, which provide an alternative to traditional home delivery, have grown in popularity in recent years, particularly with the rise of e-commerce exacerbated by the COVID-19 pandemic. These lockers streamline the delivery and pickup procedures by being strategically positioned in convenient areas like residences, office buildings, shopping malls and transit hubs. A special QR code is given to recipients, enabling them to collect their packages from dedicated lockers, guaranteeing a fast and safe delivery experience.

Finland hosts one of Europe's largest parcel locker networks, driven by consumer strong preference for out-of-home delivery options. Posti, the national postal service, manages most of these lockers (Fig. 4). Recently, other parcel locker operators like Pakettipiste and Smartmile have entered the scene. They operate on a 'carrier-agnostic' model in which lockers are designed to receive parcels from any courier, regardless of the courier company (Antonia and Willem, 2021). This unified approach enables customers to receive packages from various e-commerce platforms and courier services in a single locker which streamlines the delivery process. Additionally, the installation and maintenance costs for lockers can be shared among different courier companies and local government (Antonia and Willem, 2021). Each service point is equipped with a different number of lockers, ranging from 30 to 800 units. An IoT platform has been established to manage the two-way transmission of data between the smart lockers and Posti's system, enabling instant management of the lockers. An IoT system improves both the reliability and efficiency of parcel handling by couriers and collection by recipients. (Antonia and Willem, 2021)



Figure 4: An outdoor parcel locker operated by Posti
Source: King (2021)

Context:

The appeal of parcel lockers lies in their flexibility, which is advantageous to both carriers and recipients. Combining deliveries at centralised locations improves first-time delivery success rates and lowers delivery costs. Parcel lockers are densely located close to residential areas or along people's daily commute routes and are usually accessible 24 hours 7 days, offering flexibility for packages collection (Tang et. al., 2021). Utilisation of locker also allows online retailers to cut operation costs as multiple packages can be delivered to a single parcel locker location in one trip, while optimising delivery routes can also reduce fuel consumption and carbon emissions, making it a more environmentally friendly last mile delivery option (Tang et. al., 2021).

Parcel locker designs are tailored to accommodate parcels of different dimensions, typically featuring three compartments: small, medium, and large (Antonia and Willem, 2021). These lockers are strategically positioned over a wide range of indoor and outdoor sites, maximising accessibility for customers. Outdoor lockers, constructed to withstand harsh weather conditions, are usually situated in transportation hubs and public spaces. While indoor stations are located in supermarkets, offices, and outside parcel shops (Antonia and Willem, 2021).

Finnish consumers have generally adopted parcel lockers as their preferred delivery option with more than half of all parcels being directed to these facilities (Antonia and Willem, 2021). This preference is driven by several factors, including frequent delivery schedules, with deliveries made more than five days a week. Another key factor is the extensive coverage of lockers across various locations (Antonia and Willem, 2021). Most importantly, choosing self-collection at parcel lockers is 22-35% cheaper in delivery fee compared to home delivery (Antonia and Willem, 2021). Due to a high adoption rate, Posti having already installed 4,000 parcel locker stations by 2023, showing the remarkable success of parcel lockers in Finland.

Nevertheless, Finland's parcel locker network does encounter several challenges, such as capacity constraints during peak seasons, like Black Friday sale. Also, security concerns arise, particularly in dimly lit areas that are vulnerable to vandalism and theft (Antonia and Willem, 2021). Moreover, concerns have been raised regarding the fault-handling capability, where the locker door remains closed despite inputting the correct pin (Antonia and Willem, 2021). Taking this case study as reference, we can develop a comprehensive proposal for implementing a widespread parcel locker network in Hammersmith and Fulham.

Main Takeaways:

1. **Carrier-Agnostic Approach:** A unified system streamlines the delivery process.
2. **Cost-Effective Shared strategy:** Promotes financial sustainability and scalability through cost sharing among diverse stakeholders.
3. **Strategic Location:** Putting parcel lockers in residential areas and on regular commuting routes enhances accessibility and convenience for recipients, resulting in higher adoption rates.
4. **Versatile Design:** Customised designs that fit packages of various sizes improve usage and flexibility for carriers and customers.
5. **Reliability and Security:** By employing a smart IoT system and security camera in the lockers, technical and security concerns can be mitigated.

2.3. Railway Arches and Logistics Hubs

Background:

Economic diversity and street-level social interaction are both supported by the diversity of spatial uses, and as such there is a great potential for last-mile delivery services to be integrated with other social functions in order to produce positive social and economic externalities. In the context of London, the city's railway arches are increasingly being converted to host numerous small businesses, all of which offer a diversity of uses and opportunities for street-level activity (Froy, 2017). There are several advantages presented by the commercial and social usage of London's railway arches.

Railway arches offer a "parallel rental market", where rental prices are set by Network Rail and Transport for London, that provides more affordable and adaptable enterprise space (ibid., p. 42). This is especially significant in London, where rising property costs price out many small businesses and social enterprises, especially lower value-added economic activities, such as small-scale manufacturing (ibid., pp. 41-42). As such, railway arches serve as an incubator for a diverse range of smaller-scale, more community-oriented, commercial and social enterprises. A local last-mile delivery hub would synergise well with these small enterprises, serving as their logistics hub.



Figure 5: Visualisation of mixed-use railway arches

Source: Repurpose the City (2020)

Context:

London's railway arches offer a unique example of a 'industrial high-street', given its continuous facade, that allows a multitude of diverse small businesses and community services to exist alongside one another and re-activate former industrial areas (ibid., p. 44). Railway arches, unlike industrial estates,

tend to be found cutting through residential areas, and as such are not segregated from them. This provides a significant opportunity to embed a commercially and socially active line of railway arches in the heart of a community, providing greater commercial opportunities and stimulating local economic activity (ibid., p. 44). Moreover, the close proximity of enterprises along railway arch lines encourages a “culture of circulation”, where the movement of people across a smaller space stimulates business and social interaction (ibid., p. 45). This entails both increased cooperation and communication between local enterprises (aiding with innovation, the sharing of resources and growth), and increased social interaction between people visiting the railway arches (ibid., pp. 43-46). Providing a community-facing last-mile delivery hub would integrate well with the diverse social and commercial functions of railway arches, combining logistics, package pickup, and social mixing in one space.

However, a threat posed to the social and commercial potential of railway arches comes in the form of rising rents, speculation, and gentrification. For example, protests erupted in Brixton where plans to inflate rents coinciding with redevelopment of railway arches there threatened to “kill the backbone of Brixton” (ibid., p. 47). Rises in rents prices out the diverse range of small commercial and social enterprises that find refuge in the parallel rental market these arches offer, as well as threatening to replace local small-scale firms with new chain stores (ibid., p. 47). As such, local authorities should take steps to control railway arch rents, such as taking long leases on the arches to offer them to enterprises at affordable rates (ibid., p. 47).

Main Takeaways:

1. **Local economic and social vibrancy:** Railway arches provide the conditions for diverse local economic activity and social activity.
2. **Re-activation of industrial space:** Filling railway arches introduces commercial and social activity in otherwise unused areas.
3. **Proximity-based benefits for local enterprises and social interaction:** Increase chance of business and social interaction encourages innovation, resource sharing, commercial activity, and community building.
4. **Threat of rising costs and rents:** Railway arches are under threat of speculative rental rises and may contribute to gentrification unless regulated.

2.4. Jitu Neighbour

Background:

Jitu Neighbor is a pioneering initiative by Jitu Express in China, that combines last-mile delivery hubs with local businesses to provide a "one-stop service" solution. This integration offers services such as parcel collection and dispatch while also facilitating the sale of food and essential items within the community, also providing employment opportunities for locals, further benefiting the community.

Context:

Jitu Neighbor hubs operate on a revenue model primarily based on parcel-related activities. Parcel collection fees range from 0.4 to 0.8 RMB per item, and parcel dispatch generates 3 to 5 RMB per item, with rates negotiated with couriers or outlets. Beyond their core operations, these hubs also enhance the customer experience by integrating parcel pickups with grocery retail and other services, so merging everyday shopping essentials into their offerings (Beijing Business Daily, 2023). These activities not only support the hub's basic income but also attract significant foot traffic, which benefits adjacent local businesses, boosting revenues through additional sales in supermarkets, group purchases, and convenience services (Jetlinli.com, 2021).



Figure 6: Jitu Neighbor Hub
Source: Jetlinli.com (2021)



Figure 7: Jitu Neighbor Hub
Source: Jetlinli.com (2021)

Jitu Neighbour hubs are strategically located in areas with high occupancy and a younger demographic, typically at community centres or main entrances. The initial setup only requires an investment of about 1,000 to 3,000 RMB (about 111 GBP -333 GBP), covering rent, equipment such as shelves and printers, and minimal renovations to accommodate the operational needs (Jetlinli.com, 2021).

The synergy between last-mile delivery and local commerce can enhance the customer experience by merging essential shopping with parcel pickup services, thus optimising space utilisation and enhancing community convenience (Bai, 2023). This model can empower local merchants and shops by providing them with a broader platform. For instance, Jitu Neighbour has strategically partnered with numerous companies in China, including Alibaba and Tiktok, as well as major logistics platforms like China Post, ZTO Express, and SF Express. These collaborations streamline operations at the logistics hubs by leveraging direct partnerships to simplify processes, reduce the burden of managing extensive logistics tasks, and minimise packaging waste.

Main Takeaways:

- 1. Use of Commercial Space:** Efficiently using existing commercial spaces to host delivery hubs reduces the need for constructing new structures and maximizes the utility of underused properties. Identify underutilized areas within local businesses like Westfield and smaller retail stores that can be converted into delivery and pick-up points.
- 2. Integration of Local Shopping and Parcel Collection:** Logistics integration with existing businesses drives foot traffic, which can increase sales and visibility for these businesses. Form partnerships with local businesses where both the delivery service and the local retailers benefit from increased customer engagement and sales.
- 3. Accessibility and Convenience:** Set up user-friendly, accessible pick-up points in locations that are already part of the daily lives of the community members, such as grocery stores, malls, and other high-traffic areas, contributing to a reduction in carbon emissions.
- 4. Use of Technology:** Deploy a unified logistics management system that allows for real-time tracking of parcels, automated notifications to customers when parcels are ready for pick-up, and efficient route planning for delivery options.

2.5. Rider-Centric Last Mile Delivery – Sparqle

Background:

Sparqle, a delivery platform based in Amsterdam, sets itself apart from the increasingly saturated market by combining industry-leading delivery technology with rider-focused decisions. Sparqle says they are a “riders-centric platform”. This philosophy is clearly a response to the rise of the gig economy, where employees often face job insecurity and a lack of protection. Sparqle offers both freelance arrangements and fixed contracts. The company claims to pay roughly 40% more than other delivery apps. The technology side of Sparqle is interesting because part of their claim is that the delivery algorithm will create routes for drivers that minimise distance. Additionally, Sparqle wants to build one cohesive ecosystem – drivers and partnered businesses both sign onto the app. Sparqle allows for sustainable deliveries of goods but also are sustainable in the economic sense, by allowing delivery drivers access to higher pay, compensation etc.

Context:

Sparqle is a case study worth noting because it showcases the demand for businesses who have a clear sustainability ethos in mind. On the 23rd of February 2024, it was announced that Sparqle would receive another round of funding, amounting to 1.2 million euros, with the hopes of expanding into other European countries (Tech.EU, 2024). This round of funding was partly made possible by the Graduate Entrepreneur Fund, which supports companies founded by graduates of Rotterdam University and TU Delft. The recent infusion of capital into Sparqle is a vote of confidence from investors and businesses focused on sustainability.

However, a potential drawback for Hammersmith and Fulham’s sustainability agenda is that an infusion of outside capital into Sparqle will only serve to increase the impetus for Sparqle to become profitable. The commercial demands placed on Sparqle, as a company with outside investors, may result in a watering down of their sustainability and rider-centric ambitions. It is worth noting, that as start-up, Sparqle has yet to fully demonstrate that it can be profitable while living up to sustainability standards.

Main Takeaways:

1. **Investment Potential:** Companies showing a mindset oriented towards sustainability are increasingly valued and rewarded by investors. This trend is evidenced by Sparqle’s success in raising funding in several rounds.
2. **Effective use of technology:** Sparqle’s application and algorithm for reducing delivery times, can serve the agenda of sustainability in last-mile deliveries. Sparqle as a case study demonstrates the possibility of viewing technology as something that can be moulded and shaped to suit human needs.
3. **Equitable pay for drivers:** Sparqle shows that delivery drivers can and should be compensated fairly for their labour.
4. **Potential for Public-Private partnership:** Sparqle provides an example of a sustainability-oriented partners who understand what a council like Hammersmith and Fulham are aiming to achieve.

2.6. Case Study – Government Regulations & Policy:

The functioning of a sustainable last-mile delivery initiative must be embedded within a facilitatory regulatory and policy framework, without which the effectiveness of its implementation would be impeded in multiple ways. The hurdles faced by an implementation of last-mile delivery largely come in the form of the following issues: firstly, the financial barrier posed by the costs of e-vehicles and cargo bikes; secondly, a fragmented and inefficient competitive last-mile delivery sector; and thirdly, competition for street-space and negative environmental externalities posed by fossil-fuel based vehicles and forms of delivery. An effective regulatory framework must tackle all three of these issues to facilitate a sustainable last-mile delivery initiative.

An example of good regulatory practice can be found in the city of Suzhou, China, where official bodies of the city's government published a series of guidelines and compulsory policies to facilitate a standardised and sustainable last-mile delivery system. Between the years of 2017-2020, several policy-documents were introduced, the most significant of which are: firstly, subsidies for the acquisition of e-vehicles, cargo bikes, and sustainable packaging; and secondly, governmental support for the integration of a joint parcel delivery and infrastructure system (GIZ, 2021: pp. 5-7, 14-15).

Subsidies for e-vehicles, cargo bikes, and sustainable packaging produced several energy saving and emission reduction outcomes. Subsidised implementation of electronic waybills resulted in the saving of 1.73 billion pieces of A4 paper – the equivalent of 52,900 trees, 75.54 cubic meters of water, and 12,369 tons of carbon emissions saved (ibid., p. 31). Circular packaging has been implemented through subsidisation, resulting in a 91.9% utilisation rate of reusable transit bags and a 96.4% reduction year-on-year of disposable plastic container bags, saving 67.28 million disposable plastic bags annually (ibid., p. 31). Moreover, subsidies for the acquisition of e-vehicles have resulted in a 100% transition to standardised electric tricycle carbo-bikes in last-mile delivery, achieving “zero emissions” at this point of the delivery process (ibid., p.24). Additionally, a supplementary regulation restricted access to certain areas of the city to fossil fuel trucks, which has the effect of both reducing negative environmental externalities from fossil fuel emissions and reducing street-congestion – freeing up space for e-cargo bikes. Though not specified in the case study, such a regulation may also push delivery providers to transition away from fossil-fuel trucks to the adoption of more sustainable modes of logistical vehicles, aided by the incentives and support offered through institutional financing.

The government's push to integrate last-mile delivery and infrastructure systems greatly reduced the costs of warehousing and logistics operation by standardising and combining a previously fragmented delivery sector, eliminating unnecessary duplication and streamlining the delivery process (ibid., pp. 15-17). Within this system, multiple parcel delivery enterprises still exist and compete with one another, but function within a shared and coordinated infrastructural network (ibid., pp. 32). This integrates the advantages of competition to improve delivery and cost efficiency with coordination, infrastructural streamlining, and the sharing of resources.

As can be seen from the case of Suzhou, a regulatory framework can significantly aid in the development of a successful sustainable last-mile delivery system. The city achieved the first steps of its goal in realising a last-mile delivery system that reduces fossil fuel emissions and resource consumption while improving operational efficiency. With a combination of compulsory policies, incentives, and support, it is evident that delivery providers are willing to transition to realise a more sustainable and collaborative last-mile delivery system.

Main Takeaways

- 1. Institutional financing to encourage sustainable transition:** Subsidies are a powerful incentive to encourage rapid sustainable transition.
- 2. Multiple avenues for sustainable transition:** In last-mile delivery, transportation is not the only avenue for sustainability, but also includes other elements, such as packaging.
- 3. Supplementary traffic regulations:** Controlling street congestion facilitates the decarbonisation of logistics and increased efficiency of last-mile delivery systems.
- 4. Institutional support for integrated last-mile delivery sector:** Institutional initiative is required to push last-mile delivery providers to de-fragment the sector and integrate services.

2.7. An Alternative Approach

Considering the above takeaways from international examples and challenges present within H&F, the following section of the report will provide development assessments for the three proposed hub types, i.e. small, medium, and community-facing hubs. Following this, policy recommendations we believe are necessary to facilitate implementation will be provided.

3. DEVELOPMENT ASSESSMENT

3.1. Small Hub: Launch of Extensive Parcel Locker Network

In response to the climate and ecological emergency declared by H&F Borough Council, our proposal aims to establish an extensive network of small hubs by 2030. These will serve as centralised pick-up points for parcels within neighbourhoods, encouraging residents to transition from conventional home delivery to self-collection. The primary objective is to develop a sustainable and efficient last-mile delivery system that reduces the demand for individual delivery vehicles, thereby decreasing carbon emissions and congestion. Unlike our other recommendation, the community hub, which operates within set hours, small hubs utilise parcel lockers for parcel collection which offer round-the-clock access, operating 24/7. By strategically positioning them within the borough, we can create a dense network of small hubs, enhancing convenience for residents. With no staff presence required, parcel lockers cater to customers who seek instant and convenient parcel collection. Additionally, consolidating multiple deliveries to a single locker station saves time and resources compared to door-to-door delivery, ultimately reducing costs for courier companies. It is evident that there are strong incentives for both customers and courier companies to utilise the parcel locker network.

Key Features:

First, our proposal aims to establish a comprehensive network of small hubs, ensuring that residents can access a locker station within a 10-minute walk. In the first phase of installation, lockers will be positioned along commuting routes, particularly near Tube and overground stations (Figure 8), to ensure accessibility for all residents.

To optimal ease of use, we suggest utilising QR code technology to create a user-friendly pick-up procedure which users can easily retrieve their parcels with a quick scan. Also, installing Smart IoT Lockers with two-way data transmission abilities, referring to the Posti parcel network in Finland (Antonia and Willem, 2021). It minimises service disruptions by enabling remote fault detection and resolution. Thus, operational efficiency and fault management can be enhanced. Drawing from the Finnish model, each service point can contain lockers in three sizes: large, medium, and small, subject to demand (Antonia and Willem, 2021). To determine an optimal number of lockers at each service point, various factors including population density, purchasing power of the neighbourhood, and historical parcel delivery data should be considered (Lai et. al., 2022).

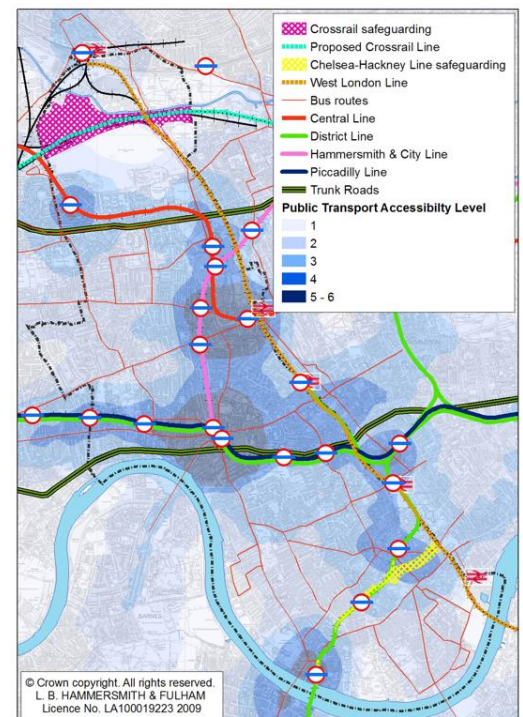


Figure 8: Map showing the spatial distribution of tube and overground stations within the borough.
Source: H&F Council (2016)

During peak periods like holidays, mobile smart parcel lockers can be arranged strategically to manage higher parcel volumes, assuring on-time deliveries, and easing congestion (Figure 9) (Schwerdfeger & Boysen, 2020).



Figure 9: A mobile parcel locker.
Source: Schwerdfeger & Boysen (2020)

In Finland's case study, consumers expressed concern about safety and security, with a focused on two key areas: package safety and person safety during pickup. Consumers who received higher-value packages expressed hesitation about using lockers (Antonia and Willem, 2021). Therefore, we suggest strategically placing lockers in well-lit and visible locations. Placing surveillance cameras and improving lighting can further prevent potential incidents of theft or vandalism, creating a secure environment for parcel collection. In addition to their practical role in facilitating parcel collection, parcel lockers can also enhance the aesthetic appeal and usability of public areas (Mitchell et al., 2022). By integrating lockers with existing street furniture, such as benches or bus stops, a cohesive urban environment that is both functional and visually pleasing can be shaped (Figure 10) (Mitchell et al., 2022).

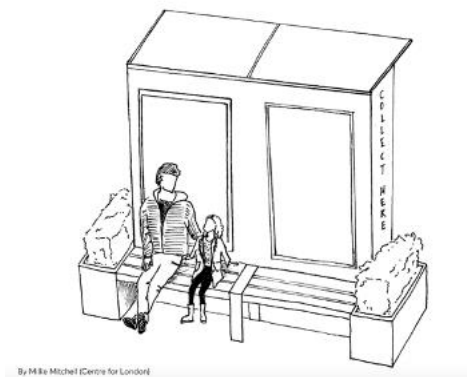


Figure 10: Visualisation of incorporating Parcel locker with a bench.
Source: Mitchell, M., Bosetti, N. and Hardiing, C. (2022)

Additionally, we propose using a 'carrier-agnostic' model within our parcel locker network, which allows packages to be delivered to our locker stations by a variety of parcel carriers, including DPD, Royal Mail, and Amazon (Lai et. al., 2022). A unified system streamlines the entire parcel delivery and collection procedure, offering convenience for customers while optimising operational efficiency for courier companies. To secure sustainable financing for this proposal, we advocate for a shared cost model that divides the cost of locker installation and maintenance between the council and courier companies. Forming partnerships with various companies would be a crucial first step. Furthermore, sustainable distribution methods can be employed for the transportation of parcels from medium hubs to parcel lockers. Parcel distribution can be minimised to once or twice daily to reduce carbon emissions and alleviate traffic congestion within the borough. To align with the Parcels Not Pollution initiative, the council may provide subsidies to encourage the usage of electric vehicles and cargo bikes. Detailed proposal will be discussed in below section.

3.2. Medium Hub

Building on the network of small hubs proposed for the H& F borough, our strategy includes developing two medium-sized logistics hubs inspired by the successful Pimlico Logistics Hub model. These hubs will act as central nodes within the broader network and serve as critical aggregation points where parcels are pre-sorted before being dispatched to the network of small-size hubs and community-facing hubs. It will also focus on reducing fossil fuel dependency, easing traffic congestion, and contributing to climate change mitigation.

Hurlingham Business Park

Located near the southern boundary of the Hammersmith & Fulham borough, Hurlingham Business Park is strategically positioned near a diverse array of company occupiers. The location's proximity to Hurlingham Harbour, Thames River offers opportunities for carrying goods for urban last-mile deliveries.

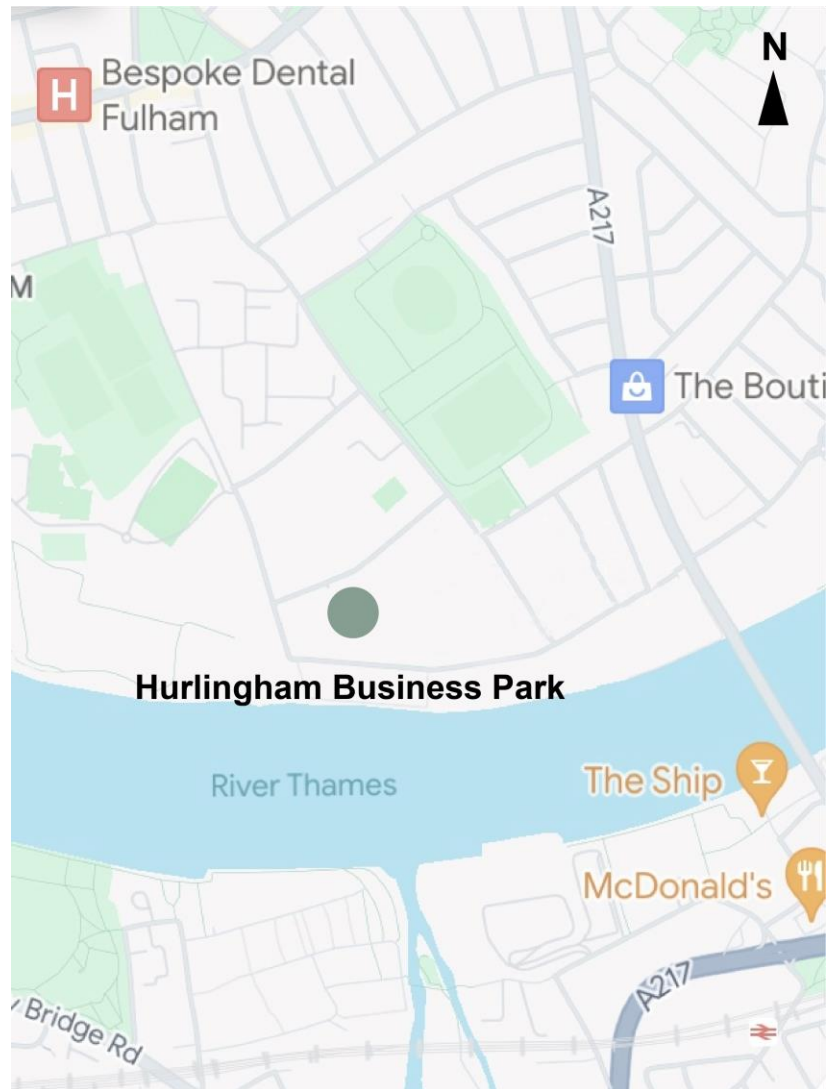


Figure 11: Location of Hurlingham Business Park
Source: Google Maps (2024)

However, there are notable challenges associated with this location. Its proximity to the Thames River increases the risk of flooding, which could disrupt operations and endanger stored parcels and infrastructure. Moreover, situated near residential areas, and the nearest main road A217 is typically exposed to high traffic volumes during peak hours, which could lead to delays in the dispatch and delivery of goods (Fig.13).

In Hurlingham Business Park, work areas and storage facilities are spread across various floors and are interconnected by internal staircases, enabling seamless integration and optimal use of space for logistical operations. This setup facilitates the effective accommodation of diverse logistical functions without necessitating extensive restructuring (Fig 12) (Frost Meadowcroft, 2023).



Estate Entrance

Figure 12: Entrance to Hurlingham Business Park
Source: Frost Meadowcroft (2023)



Figure 13: Locational Context of Hurlingham Business Park
Source: Frost Meadowcroft (2023)

Grove House

Another logistics hub is set to be established in the centre of Hammersmith and Fulham, near Hammersmith station and Thames River. This underutilised office space, vacant for several years, is designed to utilise a substantial property offering 63,224 sq. ft of self-contained office space. The entrance is directly link to the bike lane, onsite parking with 17 car spaces and cycle storage, which is advantageous given its proximity to bike lanes for cargo bike deliveries (Fig.14).



Figure 14: Entrance to Grove House

Source: Zoopla (2024)

This building features a raised floor with above-ground and four upper floors equipped with air-conditioning, building management systems, passenger lifts, showers and other facilities. The whole property also benefits from accessible roof terraces (Fig. 15).



Figure 15: Grove House Office

Source: Zoopla (2024)

However, the current electrical system tailored for office use may not meet the higher power demands typical in logistics operations, which require extensive use of automated systems and continuous heavy machinery operation, and enhancing the power supply to ensure stability and continuous operation may also serve an issue.

Locational Synergy

Hurlingham Business Park and Grove House are strategically designed to function in synergy, and each have a focusing point, they are situated a brief 21-minute cargo bike ride apart, which will limit contribution to congestion (Fig. 16).

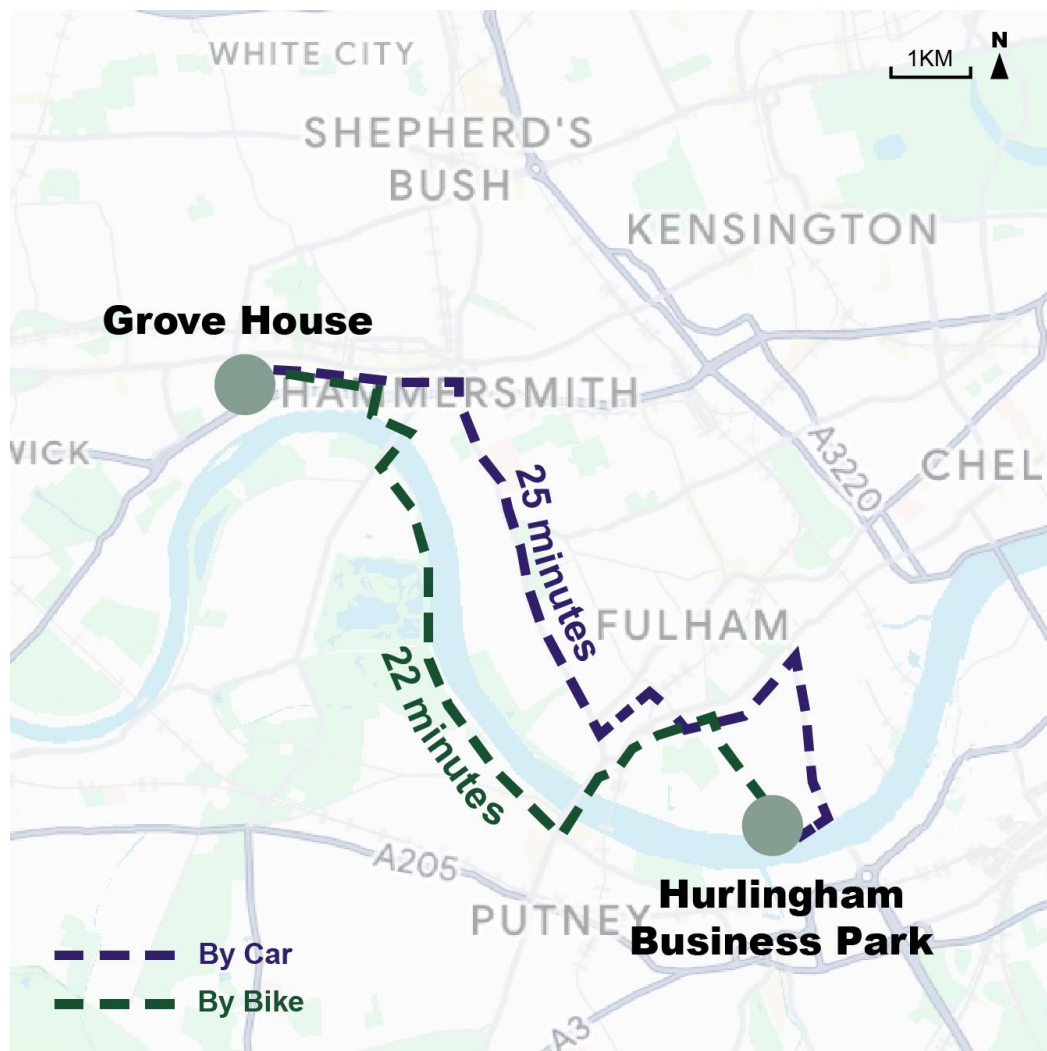


Figure 16: Distance between Medium Hubs

1. Hurlingham Business Park will specialise in the consolidation and advanced sorting of bulk deliveries, utilising state-of-the-art automated sorting systems and high-density storage solutions. It is designed to efficiently process large volumes of parcels for local distribution. This hub also supports the

maintenance of cargo bikes, which boosts the operational capabilities of Grove House, facilitating eco-friendly last-mile deliveries.

2. Grove House will transform to specifically address distribution, positioning as a crucial hub for small-size hub and community-based self-pickup points. Drawing inspiration from the Q-Park Pimlico project, it will focus on minimising noise and maximising efficiency, particularly in the storage of electric cargo bikes on the ground floor and in designated, limited outdoor parking spaces. It is also recommended that Grove House install solar panels on its rooftop to reduce reliance on non-renewable energy sources and support the building operations with green energy.

Both buildings feature multiple levels, allowing them to store parcels or goods on higher floors to mitigate flood risks. Large goods vehicles or light goods vehicles transport items to these two hubs, then they will carry them out using self-pick-up, cargo bikes, or small electric trucks. This integrated medium-size hub system will enable real-time updates on parcel status and location, ensuring seamless distribution to small-size hubs and community-based hubs throughout Hammersmith & Fulham.

3.3. Community – Facing Hubs

Community-facing hubs are designed to be spaces that are both functional and social. Community members will be able to pick up their packages but while doing so, may run into neighbours, co-workers, friends etc. Community-facing hubs integrate package pick-ups with other errands such as grabbing a coffee or doing a small grocery shop to build a sense of social cohesion. The hope is that more casual, chance encounters will take place in our proposed Community-Facing Hubs. The community-facing hubs are sustainable in the realm of carbon-emission reduction because they provide easily accessible sites for residents to pick up their parcels, hence removing a delivery vehicle from the process. But equally so, our community-facing hubs are sustainable from a social point of view. Residents are better able to take ownership of the spaces when they can participate in shaping them. We hope this will contribute to the longevity of our proposed community-facing hubs.

We have identified two potential sites for these community-facing hubs, as represented in the map below:

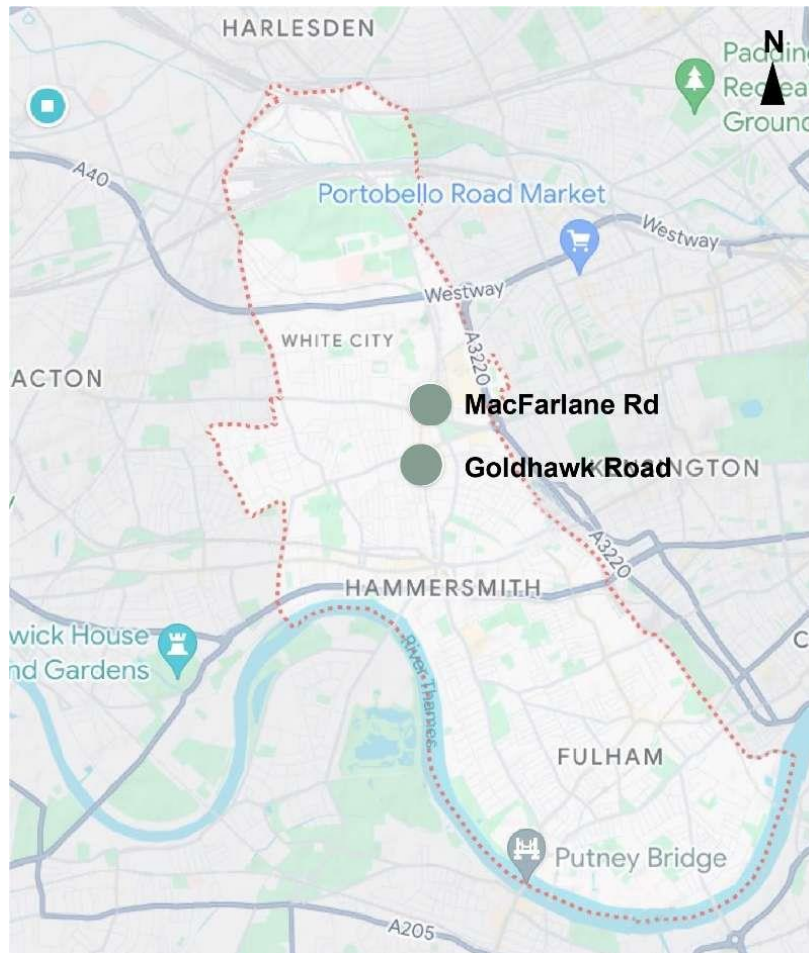


Figure 17: Map showing potential locations for Community-facing Hubs.

MacFarlane Place

The first proposed site is MacFarlane Place Arches, located directly beside MacFarlane Place, a new-build residential development. The proposed site is partially occupied by a motorcycle repair shop and a gym. Given the proximity of Westfield White City, a community-facing hub in this location offsets the presence of large multinational chains in the mall by giving local businesses a platform to showcase their products and services. A community-facing hub will allow people picking up their parcels to be more grounded in their local area. Additionally, MacFarlane Place, which is the residential development located directly beside the arches, represents large-scale change to the built environment in the area. A community-facing hub helps residents to hold onto the area's identity.



Figure 18: Rear View of MacFarlane Place arches



Figure 19: Front View of MacFarlane Place arches

Goldhawk Road

Our second proposed site several arches in Goldhawk Road, located within a short walking distance to Shepherd's Bush Station as well as the Shepherd's Bush Market, a lively produce and food market. Placing a community-facing hub in Goldhawk Road creates one-block cluster of transport, parcel collection, and whole food market. We propose that part of the community-facing hub includes an allocation for more stalls for the Shepherd's Bush Market. We hope to create a collaboration between

the community-facing hub and vendors of Shepherd's Bush Market. The community-facing hub is envisioned to be embedded within the existing community infrastructure, not as something separate. The intention behind this site is for residents to be able to arrive at their local station, pick up their parcels, and then be encouraged to explore Shepherd's Bush Market.



Figure 20: Goldhawk Road Arches

4. POLICY PROPOSALS

4.1. Partners and Governance

Given the context of fiscal constraints of local councils like H&F, the success of a sustainable last mile delivery solution in the area will likely depend on partnering with the right organisations and building a form of community governance. Councils may not be able to completely subsidise green deliveries, so this is an opportunity to involve members of the local Hammersmith and Fulham communities.

One of our key proposals in this realm is incentivizing small and medium sized local enterprises to be vendors in our proposed hubs, particularly for community-facing hubs. In policy terms, local businesses might be afforded the right of first refusal of commercial spaces in the hubs. In cases where local businesses are not able or willing to occupy these spots, larger brands will then be given the opportunity to fill the spaces.

The primary benefit of a local-first approach to vacancies is that flows of money can be kept within the community. From a social point of view, an additional hope is that locally owned businesses are more likely to facilitate a sense of social cohesion and community identity. However, for the customers in H&F, there may be impacts on pricing and affordability. Multinational chains are often able to charge lower prices. Residents of Hammersmith and Fulham, the very people this proposal aims to reach, may be priced out by local businesses.

4.2. Regulations and Policy

A solid regulatory and policy is required in order to facilitate the implementation of a sustainable last mile delivery system. As such, the Borough of H&F must take a proactive approach to providing the conditions within which this system can function and thrive. Key areas for the borough to tackle are the fragmentation of the last-mile delivery sector, incentivising sustainability transitions, and tackling precarity of last-mile delivery workers.

The last-mile delivery market in London is highly fragmented and split between a multitude of small companies, resulting in a non-cooperative and overly de-centralised system that generates inefficient duplication of services (Zhang et al., 2022: p. 1473). Moreover, a lack of coordination and the proliferation of last-mile delivery providers has resulted in an unequal distribution of workload, inefficient utilisation of resources and infrastructure, and freight traffic congestion (ibid., p. 1473). Within this unregulated system, the efficiency of last-mile delivery services would worsen, alongside the negative externalities of this inefficiency including congestion, delivery costs, pollution, and road accidents. As such, it is crucial for H&F to promote and regulation coordination and cooperation between last-mile delivery service providers while maintaining sufficient competition. A key solution would be the provision of a neutral infrastructural service and framework under a “Coopetition Scheme” (Xiao & Feng, 2010). Within this scheme, last-mile delivery providers would still maintain competition between one-

another in offering their services, but would share the same facilities, infrastructure, and coordinated workload. This reduces the negative externalities of a fragmented delivery market by balancing out the workload and efficiently rationalising infrastructure and resources, whilst also improving the competitive efficiency of each delivery service, providing an additional barrier against monopolisation of last-mile delivery services.

The rapid emergence of bicycle delivery services across the world has coincided with a concerning growth of precarious working conditions for workers in this sector. These delivery workers tend to be self-employed contractors who gain very little in the form of labour and social protection due to the nature of their work in an unregulated gig economy. This is an issue that H&F cannot overlook in the development of a sustainable last-mile delivery system. Therefore, it is highly encouraged that the borough, in its efforts to develop a more integrated 'Coopetition' last-mile delivery system, should implement a highly selective criteria for which delivery services should be integrated according to the labour relations they offer to their workers. In collaboration with trade unions, such as the *Independent Workers' Union of Great Britain (IWGB)*, the borough should form a panel of experts to review last-mile delivery services operating in London and in the borough and use these findings to determine which services should be supported. This not only ensures that the borough supports socially sustainable last-mile delivery services, but also provides a powerful incentive for other providers to fulfil these labour requirements in order to be integrated into the boroughs last-mile delivery system.

The high cost of electric vehicles, including e-cargo bikes and e-cargo tricycles, discourages existing delivery providers from transitioning their fleets from fossil fuels to electric. As such, various solutions are possible to address this issue, including the use of Private-Public Partnerships with delivery providers and the use of institutional financing in order to encourage, incentivise, and subsidise the take-up of e-vehicles and e-bikes for last-mile delivery. Collaboration is key here in order to split up the costs of equipment acquisition and retrofitting infrastructure. H&Fs existing 'Parcels not Pollution' program, for example, is a positive example of an institutional financing means that should be continued and even expanded to include heavier-duty modes of sustainable last-mile delivery. Likewise, similar mechanisms can be employed to encourage the adoption of more sustainable packaging material, to incentivise the use of re-usable delivery bags and electronic way-bills to cut down on paper and cardboard usage.

4.3. Behaviour Change

One of the biggest potential limitations for this approach to last-mile delivery lies in encouraging residents to switch from home-delivery and to increasingly engage with small and community-based hubs. Though incentives have been provided in the form of locational convenience and integration within the community, an effort must be made to engage residents within this system.

Therefore, it is suggested that adjacent to the development of these hubs, H&F council actively promote the use of community hubs and encourage residents to integrate the use of these hubs into their daily lives. This can be carried out in a similar manner to the "Love your local Highstreet" campaign,

emphasizing that these hubs are conveniently located, create an opportunity to explore the local area, and serve as a cheaper, green alternative to home delivery. In this way, residents are increasingly encouraged to view themselves as active participants in H&F vision for net zero, and to contribute to their local economy.

Furthermore, to ensure this approach to last mile-delivery benefits the local community, these hubs should promote shopping locally and uptake of active transport methods. As a result, advertising within and around these hubs should be strictly limited to the promotion of community initiatives, council news, and provide spaces for local businesses to advertise their services.

4.4. Whole Community Delivery

Within our vision, there is clear focus on urban logistics hubs as convenient sites for collecting parcels, in addition to accommodating and promoting active travel. But, as previously mentioned, this may not fully allow for people who are less able to use these hubs to be fully integrated into this system.

As a result, it is proposed that adjacent to the establishment of these hubs, a volunteer-based porter-delivery system can also be established. In a similar manner to the Parcels not Pollution scheme, residents with limited mobility are able to order parcels to a community-based hub, where they are received, and residents notified. These parcels are then collected by volunteers and taken to the resident via bike or foot. Unlike existing last-mile delivery, this system will focus on ensuring parcels are directly handed to the resident. As demonstrated within H&F 2030: Climate and Ecology Strategy (2021), there is a willingness from residents to be able to engage in flexible volunteering in order to contribute to the net zero vision, and this provides the opportunity to do so. Furthermore, this system would train volunteer couriers to actively engage with the residents they delivery to, further building social capital and demonstrating a friendly alternative to existing time-restricted delivery.

5. CONCLUSION

5.1 Challenges of application within H&F

Though the suggestions outlined in this paper are based on practical and evidence-based research and has offered robust solutions to a number of issues the borough faces, there are a number of limitations that must be recognised.

Active travel and Accessibility

As stated, one of the main focuses of these hubs is encouraging active transportation methods, such as walking and cycling. But collection-oriented approaches to last-mile delivery limits accessibility for individuals who struggle with mobility or are otherwise unable to reach hubs. While these initiatives would occur alongside existing infrastructure and the proposed whole-community delivery approach, actively integrating all community members into a new last-mile delivery system is crucial for long-term success and for empowering local people to feel a part of the H&F's low-carbon future.

Council Finances

As is the case with councils nationwide, Hammersmith and Fulham are facing an increasingly limited budget with which to meet various community needs, while simultaneously aiming to invest in a green and equitable future. Council finances is a significant element to the success of a last-mile delivery system, with multiple requirements required to be well funded to achieve long-term success. As shown above, opportunities also arise in partnering with large delivery companies such as Amazon already integrating the use of cargo bikes within their delivery systems, to fund and/or purchase the vehicles needed to facilitate this system (Amazon, 2023). But the building and establishing of such partnerships can also be costly, time-consuming, and may lead to governance structures which do not prioritise the specific needs of the local community. As a result, H&F Council must strike a balance between Public-Private Partnerships without compromising on the needs and interests of the local community in securing funding.

Energy Availability

Many of these initiatives rely on e-vehicles which in turn require EV charging facilities, in addition to the provision of charging stations for those on bike/foot. Despite having one of the highest numbers of EV charging points within West London, like the rest of the boroughs within the West London Alliance, H&F are facing an increased strain on energy provision. Though beyond the scope of this paper, long-term solutions to secure increased energy supply would be needed to maintain the sustainability of the proposed last-mile delivery system. While recent action such as the development work of the West London Local Area Energy Plan (Arup, 2023) indicates potential for an expansion of EV provision throughout the borough, the provision of the energy required continues to provide a challenge.

5.2. Summary of Findings and Proposals

This paper recommends that Hammersmith & Fulham implements a last-mile delivery system centred around the vision of Parcels, People and Planet. The recommendations laid out thus far enable the borough to implement a logistics network that promotes the gradual decoupling from wider global supply chains by anchoring the distribution and exchange of resources locally. Moreover, these recommendations emphasise a more environmentally and socially sustainable model of logistics, both offering a more efficient means of delivery as well as one that better protects drivers, stimulates local commercial activity, and strengthens community ties.

From an analysis of existing examples and case studies, it has been demonstrated that a variation of scales is necessary to establish an effective and holistic last-mile delivery chain in H&F – divided up into Medium Hubs, Small Hubs, and Community-Facing Hubs. The Medium Hubs in Hurlingham Business Park and Grove House would offer secure and centralised logistics centres from where the last-mile delivery process begins that connects to comprehensive network of strategically located Small Hubs composed of parcel lockers – enabling both direct delivery and self-pickup. Additionally, the incorporation of a number of Community-Facing Hubs integrates small logistics hubs in areas of social and commercial activity, such as converted railway arches and high streets, synergising with these activities and enterprises.

As shown by additional case studies, this network would need to be structured within a policy and regulatory framework that encourages social and environmental sustainability. Firstly, the last-mile delivery hubs should be integrated with local businesses and coincide with a campaign to encourage shopping locally, exploiting the local logistics network to enhance Hammersmith & Fulham's commercial activity and inter-connectedness. Secondly, an integration of local businesses into H&F's last-mile delivery system must also introduce community governance, ensuring that the people of the borough have agency in decision-making processes to protect local businesses and promote community interests. Thirdly, local government must act decisively to provide the regulatory and policy basis for the efficient functioning of a socially and environmentally sustainable last-mile delivery system, including the de-fragmentation of the last-mile delivery sector with an agonistic 'Coopetition' system, subsidising the transition to more sustainable modes of transport and packaging, and ensuring last-mile delivery providers offer sufficient labour protections for their workers. Lastly, H&F must be aware of enabling accessibility with the last-mile delivery system, including a flexible volunteering system to ensure package delivery to residents with limited mobility. With this diverse and holistic set of policy and regulatory approaches, a solid framework can be established around the last-mile delivery infrastructural network that provides the conditions for local commercial activity, community engagement, environmental sustainability, and a more cooperative and fairer delivery sector.

The development of a socially and environmentally sustainable last-mile delivery system, that stimulates local commercial and social activity, and one that offers safe and secure employment for workers, is both an ambitious and difficult task. As evidenced in this paper, a multi-scalar and holistic approach is fundamentally necessary, one that does not see last-mile delivery only through the lens of cost and time efficiency but also as a crucial public good for local economic and social activity in Hammersmith & Fulham, the benefits of which compound over time following initial investment and effort. This is an opportunity for the borough to buck the trend in the erosion of labour security and local economic

activity, to use the last-mile delivery system to offer a safe form of employment and a means through which the local community can distribute resources more efficiently and stimulate economic activity anchored within the community.

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